

Lecture 3 - Quality Attributes: Case Study Presentations in Week 4

Topic: Quality Attributes, Scalability, Throughput, and Security
Format: Each Group: 5-10-Minute Live Presentation & Discussion

Case Study Presentations

Group 1: The "Morning Rush" Architects

Scenario: The Midland Hotel is overhauling its lobby system. The front desk staff are constantly stressed, especially during the 9:00 AM check-out rush and the 3:00 PM check-in wave. You have been hired to define the system's quality standards.

Presentation Prompts:

1. **QA Identification:** What are the specific Quality Requirements relevant to the *Check-In* and *Check-Out* tasks?
2. **Compare & Contrast:** What quality requirements are similar between these two tasks, and which ones are different? (Hint: Think about what the guest is doing/feeling during each).
3. **Verifiable Spec:** Write one highly precise, verifiable requirement for the Check-Out task.
4. **The Big Picture:** Taking a step back and looking at the *entire* Hotel Information System, what are the top 3 to 5 critical quality attributes?

Group 2: The Infrastructure Engineers

Scenario: A major online booking engine, "GlobalStays," works perfectly for 11 months of the year. However, every Black Friday, the site completely crashes, losing millions of dollars in bookings. Management wants to know why the system can't handle the traffic.

Presentation Prompts:

1. **Throughput Analysis:** Explain to management what "Throughput" means. Why did the previous architects fail by designing the system based on "average throughput" instead of "peak throughput"?
2. **The Scaling Solution:** To fix this, the system needs Scalability. Explain the difference between "Scaling Up" (Vertical) and "Scaling Out" (Horizontal).
3. **Recommendation:** Which scaling method would you recommend for a modern web application that experiences massive, unpredictable holiday spikes?

Group 3: The Cybersecurity Consultants

Scenario: SecureBank recently updated its internal staff portal. To maximize security, they instituted 12-character rotating passwords, biometric scans, and a 3-minute inactivity timeout. Now, bank tellers are complaining that the system is completely unusable and goes offline constantly due to server load from encryption checks.

Presentation Prompts:

1. **AuthN vs AuthZ:** Clearly explain the difference between Authentication (AuthN) and Authorization (AuthZ) in this banking system.
2. **The Trade-off:** How does maximizing the quality attribute of **Security** directly negatively affect the quality attribute of **Availability**? Explain the mechanical/human reasons why this trade-off occurs.

Group 4: The Tech-Debt Analysts

Scenario: A tech startup has a wildly popular mobile app. The users love it (it's fast and looks great). However, the developers are miserable; the codebase is a "spaghetti" mess, and adding a new feature takes three months.

Presentation Prompts:

1. **Internal vs. External:** Software qualities are categorized into Internal and External. Explain both categories. Based on the scenario, which category is succeeding and which is failing? Give examples.
2. **General Attributes:** Give a quick summary of the 6 general software quality attributes (Availability, Modifiability, Performance, Security, Testability, Usability) and identify which specific attribute this startup is currently failing at.